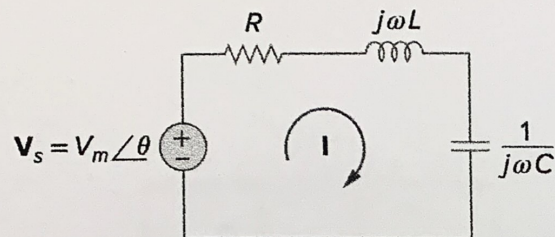


# Practice Problem 14.7



A series-connected circuit has  $R = 4 \Omega$  and  $L = 25 \text{ mH}$ . (a) Calculate the value of  $C$  that will produce a quality factor of 50. (b) Find  $\omega_1$ ,  $\omega_2$ , and  $B$ . (c) Determine the average power dissipated at  $\omega = \omega_0$ ,  $\omega_1$ ,  $\omega_2$ . Take  $V_m = 100 \text{ V}$ .

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**Answer:** (a)  $0.625 \mu\text{F}$ , (b)  $7920 \text{ rad/s}$ ,  $8080 \text{ rad/s}$ ,  $160 \text{ rad/s}$ , (c)  $1.25 \text{ kW}$ ,  $0.625 \text{ kW}$ ,  $0.625 \text{ kW}$ .

$$\textcircled{1} Q = \frac{\omega_0 L}{R} = 50 \Rightarrow \omega_0 = \frac{50 \times 4}{25 \text{ m}} = 8 \text{ k} \quad \left. \begin{array}{l} \omega_0 = \frac{1}{\sqrt{LC}} \end{array} \right\} \Rightarrow C = \frac{1}{\omega_0^2 \cdot L} = \frac{1}{(8 \text{ k})^2 \cdot 25 \text{ m}} = 0.625 \mu\text{F}$$

$$\textcircled{2} Q = \frac{\omega_0}{B} \Rightarrow B = \frac{8 \text{ k}}{50} = 0.16 \text{ k rad/s} \therefore \omega_1 = \omega_0 - \frac{B}{2} = 7.92 \text{ k rad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 8.08 \text{ k rad/s}$$

$$\textcircled{3} \text{ 谐振频率处. } P|_{\omega_0} = \frac{1}{2} \frac{V_m^2}{R} = \frac{1}{2} \frac{100^2}{4} = 1.25 \text{ kW}$$

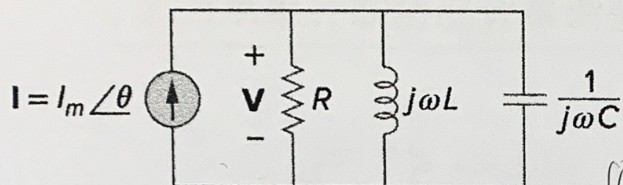
$$P|_{\omega_1} = P|_{\omega_2} = \frac{1}{2} P|_{\omega_0} = 0.625 \text{ kW}$$



## Practice Problem 14.8

A parallel resonant circuit has  $R = 100 \text{ k}\Omega$ ,  $L = 50 \text{ mH}$ , and  $C = 2 \text{ nF}$ . Calculate  $\omega_0$ ,  $\omega_1$ ,  $\omega_2$ ,  $Q$ , and  $B$ .

**Answer:** 100 krad/s, 97.5 krad/s, 102.5 krad/s, 20, 5 krad/s.



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$$① \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \text{ m} \cdot 2 \text{ n}}} = 100 \text{ k rad/s}$$

$$② \quad Q = \frac{R}{\omega_0 L} = \frac{100 \text{ k}}{100 \text{ k} \cdot 50 \text{ m}} = 20$$

$$③ \quad Q = \frac{\omega_0}{B} \Rightarrow B = \frac{100 \text{ k}}{20} = 5 \text{ k rad/s}$$

$$④ \quad \omega_1 = \omega_0 - \frac{B}{2} = 97.5 \text{ k rad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 102.5 \text{ k rad/s}$$



Calculate the resonant frequency of the circuit in Fig. 14.29.

**Answer:** 173.21 rad/s.

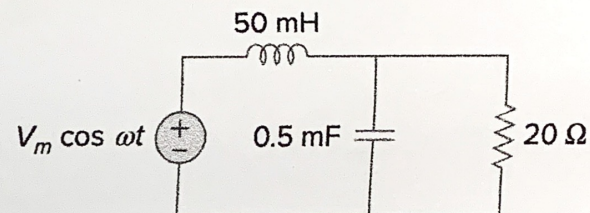
非简单串并联情况计算谐振频率

$$\begin{aligned} Z &= j\omega L + \frac{1}{j\omega C + \frac{1}{R}} \\ &= j\omega L + \frac{R}{1 + j\omega RC} \\ &= j\omega L + \frac{1 - j\omega RC}{1 + \omega^2 R^2 C^2} \cdot R \end{aligned}$$

令虚部 = 0

$$\Rightarrow \omega L = \frac{\omega R^2 C}{1 + \omega^2 R^2 C^2} \Rightarrow \omega = \frac{\sqrt{\frac{R^2 C}{L} - 1}}{RC} = 173.205 \text{ rad/s}$$

## Practice Problem 14.9



**Figure 14.29**  
For Practice Prob. 14.9



## Practice Problem 14.11

Design a band-pass filter of the form in Fig. 14.35 with a lower cutoff frequency of 20.1 kHz and an upper cutoff frequency of 20.3 kHz. Take  $R = 30 \text{ k}\Omega$ . Calculate  $L$ ,  $C$ , and  $Q$ .

**Answer:** 23.87 H, 2.6 pF, 101.

### 带通滤波器

$$\omega_1 = 20.1 \text{ k} \cdot 2\pi$$

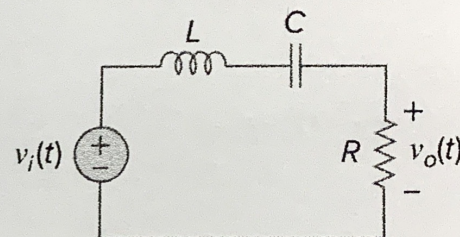
$$\omega_2 = 20.3 \text{ k} \cdot 2\pi$$

$$\therefore \omega_0 = \frac{\omega_1 + \omega_2}{2} = 20.2 \text{ k} \cdot 2\pi$$

$$Q = \frac{\omega_0}{B} = \frac{20.2 \text{ k}}{0.2 \text{ k}} = 101$$

$$Q = \frac{\omega_0 L}{R} \Rightarrow L = \frac{101 \times 30 \text{ k}}{2\pi \cdot 20.2 \text{ k}} = 23.873 \text{ H}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow C = \frac{1}{\omega_0^2 L} = 2.6 \text{ pF}$$



**Figure 14.35**  
A band-pass filter.